A Macro-Level Bandwidth Model for Monolithic 3D Memory

3d_memory — design-space exploration tool

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Abstract

We give the analytical kernel behind `3d_memory`, a macro-level design-space-exploration tool for monolithic 3D memories at the CACTI/DESTINY/NVSim altitude. Given a fixed storage capacity C, a fixed footprint A, a fixed layer budget L, and a peak-temperature constraint $\Delta T_{\max}$, the tool searches 3D organizations that *maximize the internal array-supply bandwidth.* The central object is a throughput-bound cycle time, a bandwidth identity linear in the number of independent peripheral sets, and a volume-packing constraint that ties capacity, cell density, and the layer partition together. Routing, PDN, and the thermal boundary condition are deferred (Sec. 6).

1 Problem statement

Given a fixed capacity C, a fixed footprint A, a fixed number of monolithic tiers L, and a peak-temperature budget $\Delta T_{\max}$; *find* the 3D memory organization (array geometry, degree of peripheral sharing, sense-amp stacking, and tier assignment of circuit classes) that *maximizes* the internal array-supply bandwidth BW.

Two named 3D levers act on this objective:

1. **Peripheral stacking** — tucking sense amps, decoders, and selectors on tiers under/over the array (CUA / Choi-style), freeing footprint and relieving pitch.
2. **Multi-tier bitcells** — placing different devices of a single cell on different tiers, raising areal cell density.

2 Nomenclature

Symbol	Meaning
$C,\ A,\ L$	storage capacity, footprint (area/layer), layer budget
$\Delta T_{\max}$	peak temperature rise allowed (thermal governor)
a_{cell}	areal footprint of one stored bit (cell density^{-1})
$N_{\text{arr,indep}}$	number of *independent* peripheral sets (arrays or blocks of shared arrays)
$N_{\text{arr,share}}$	number of arrays time-multiplexed onto one shared peripheral set
$N_{\text{arr,tot}}$	total array count, $N_{\text{arr,tot}} = N_{\text{arr,indep}} \cdot N_{\text{arr,share}}$
$N_{\text{SA}},\ N_{\text{dec}}$	number of sense amps, number of decoders
b_{acc}	bits delivered per access (sense width)
$N_{\text{BL,arr}},\ N_{\text{WL,arr}}$	bitlines, wordlines per array
t_{cycle}	steady-state cycle time (throughput bound)
$t_{\text{dec}}, t_{\text{WL}}, t_{\text{BL}}, t_{\text{SA}}$	decode, wordline, bitline-develop, and sense (incl. restore) times
t_{switch}	selector switching time for time-muxing shared arrays
$\text{Vol}_{\text{arr}}, \text{Vol}_{\text{SA}}, \text{Vol}_{\text{dec}}, \text{Vol}_{\text{sel}}$	volumes of one array, sense amp, decoder, selector network
$L_{\text{arr}}, L_{\text{periph}}$	layers holding bits vs. layers holding periphery
A_{SA}	footprint of one sense amp (offset-limited)
K	column-multiplexing degree
$w_{\text{SA}},\ p_{\text{BL}}$	sense-amp pitch, bitline pitch
u	areal utilization of a peripheral tier

3 The kernel

3.1 Cycle time (throughput bound)

The steady-state cycle time is set by the slowest *un-hideable* resource occupancy. The shared decoder and sense-amp occupancies are floors that cannot be pipelined away; the array develop latency $t_{\text{WL}} + t_{\text{BL}}$ can be hidden by interleaving $N_{\text{arr,share}}$ arrays onto the shared periphery:

$$t_{\text{cycle}} = \max\left(t_{\text{dec}},\ \ t_{\text{SA}} + t_{\text{switch}},\ \ \frac{t_{\text{dec}} + t_{\text{WL}} + t_{\text{BL}} + t_{\text{SA}} + t_{\text{switch}}}{N_{\text{arr,share}}} \right). \tag{1}$$

t_{dec} is the critical decoder-stage delay; t_{SA} *includes restore* for destructive-read cells. This last inclusion silently encodes the technology dependence of sharing: for a destructive cell (DRAM) t_{SA} floors near t_{RC} and the shared term can never dominate, so interleaving buys nothing; for a non-destructive cell (gain cell, NVM) t_{SA} is small and interleaving pays.

3.2 Bandwidth

Internal array-supply bandwidth is the per-access word width divided by the cycle time, replicated across every *independent* peripheral set:

$$\boxed{\text{BW} = \frac{b_{\text{acc}}}{t_{\text{cycle}}} N_{\text{arr,indep}}} \qquad \text{with} \qquad b_{\text{acc}} \le N_{\text{BL,arr}}. \tag{2}$$

The cap $b_{\text{acc}} \le N_{\text{BL,arr}}$ states that no access can sense more bits than there are bitlines (whole-row-activate limit). Sharing raises $N_{\text{arr,tot}}$ and hides latency but never appears in (2): it adds capacity, not bandwidth.

3.3 Accounting identities

$$N_{\text{arr,tot}} = N_{\text{arr,indep}} \cdot N_{\text{arr,share}}, \qquad N_{\text{SA}} = N_{\text{arr,indep}} \cdot b_{\text{acc}}, \qquad N_{\text{dec}} = N_{\text{arr,indep}}. \tag{3}$$

There is one decoder per independent set, and the sense-amp count is the total delivered word width.

3.4 Volume (packing identity)

Consumed volume equals the sum of every packed component — an equality, not a bound, so that no volume goes unaccounted:

$$\text{Vol} \;=\; \text{Vol}_{\text{arr}} \cdot N_{\text{arr,tot}} \;+\; \text{Vol}_{\text{SA}} \cdot N_{\text{SA}} \;+\; \text{Vol}_{\text{dec}} \cdot N_{\text{dec}} \;+\; \text{Vol}_{\text{sel}}(N_{\text{arr,share}}). \tag{4}$$

The selector term $\text{Vol}_{\text{sel}}(N_{\text{arr,share}})$ is the volume of the network that time-multiplexes $N_{\text{arr,share}}$ arrays onto one shared peripheral set; it grows with the sharing degree and is what makes sharing cost area, not just time.

3.5 Scaling laws

$$t_{\text{WL}} = k_{WL}^{\text{wire}}\, N_{\text{BL,arr}}^2 + k_{WL}^{\text{cell}}\, N_{\text{BL,arr}}, \qquad t_{\text{BL}} = k_{BL}^{\text{wire}}\, N_{\text{WL,arr}}^2 + k_{BL}^{\text{cell}}\, N_{\text{WL,arr}}, \tag{5}$$

$$\text{Vol}_{\text{dec}} \propto N_{\text{WL,arr}} \cdot N_{\text{BL,arr}} \propto \text{Vol}_{\text{arr}}, \qquad \text{Vol}_{\text{SA}} \propto \frac{1}{\text{margin}^2}. \tag{6}$$

Each develop time separates into a *wire* self-RC term and a *cell* loading term. The wire term is the distributed-line delay $\frac{1}{2} rcL^2$ (the CACTI-6 $n(n{+}1)rc/2$ form) with L set by the number of cells along the line, hence quadratic; its coefficient $k^{\text{wire}} = \frac{1}{2} rc$ is fixed by the BEOL metal stack and node ($\times$ tier count in monolithic 3D) and is therefore *shared* across cell technologies on the same wire stack. The cell term is the storage/access-device capacitance charged through the access resistance, roughly linear in the cells on the line, and is the knob set *per technology* (k^{cell}: DRAM storage cap, gain-cell read-FET gate cap, memristor device cap, …). Together they are what makes large unsegmented arrays expensive. Decoder volume tracks the array it drives; sense-amp volume follows the Pelgrom offset law $A_{\text{SA}} \propto 1/\text{margin}^2$, so tighter sensing margins cost quadratic area.

4 Derived structure

4.1 Layer partition

Capacity and footprint fix how many tiers must hold bits; the rest are free for periphery:

$$L_{\text{arr}} = \left\lceil \frac{C\, a_{\text{cell}}}{A} \right\rceil, \qquad L_{\text{periph}} = L - L_{\text{arr}}, \qquad N_{\text{SA}} \le \frac{A\, L_{\text{periph}}\, u}{A_{\text{SA}}(\text{margin})}. \tag{7}$$

Bandwidth lives in L_{periph}. Hence **cell density is a quantified indirect bandwidth lever**: shrinking a_{cell} (e.g. by making the cell 3D / multi-tier) lowers L_{arr} and frees peripheral tiers for more sense amps — worthwhile iff the freed tiers buy more N_{SA} than the added cell complexity costs in Vol_{arr}.

4.2 Optimal sharing degree

Interleaving helps only until the develop term in (1) drops to the floor; beyond that, more sharing adds $\mathrm{Vol}_{\mathrm{sel}}$ and $N_{\mathrm{arr,tot}}$ for no bandwidth:

$$N^{\star}_{\mathrm{arr,share}} = \left\lceil \frac{t_{\mathrm{dec}} + t_{\mathrm{WL}} + t_{\mathrm{BL}} + t_{\mathrm{SA}} + t_{\mathrm{switch}}}{\max(t_{\mathrm{dec}},\ t_{\mathrm{SA}} + t_{\mathrm{switch}})} \right\rceil. \tag{8}$$

4.3 Width / mux / pitch coupling

The sense width is set by the column-mux degree, which is floored by the sense-amp pitch; stacking n sense-amp tiers relieves that pitch:

$$b_{\mathrm{acc}} = \frac{N_{\mathrm{BL,arr}}}{K}, \qquad K \geq \left\lceil \frac{w_{\mathrm{SA}}}{p_{\mathrm{BL}}} \right\rceil \Big/ n. \tag{9}$$

So the width lever, the mux degree, the sense-amp pitch, and sense-amp tier stacking are a single coupled knob.

5 Allocation recipe

Given $(C, A, L,$ cell/tier tech), the optimizer walks the levers in dependency order:

1. **Cell density** — fixes L_{arr} via (7), freeing L_{periph}.
2. **Array geometry** — fixes the t_{cycle} floor (1) and $\mathrm{Vol}_{\mathrm{arr}}, A_{\mathrm{SA}}$.
3. **Sense-amp stacking + mux degree** — raises b_{acc} toward $N_{\mathrm{BL,arr}}$ (9).
4. **Share to $N^{\star}_{\mathrm{arr,share}}$** — reaches the t_{cycle} floor at minimum cost (8).
5. **Fill L_{periph}** — spend remaining peripheral area on N_{SA} (7), maximizing (2).

6 Scope and deferrals

Phase 1 is a *packing* model: how much bandwidth-producing hardware fits in L tiers of area A at capacity C. Two assumptions justify deferring routing: (i) **distributed vertical bottom-exit** — each independent array exits its bits through a local per-array vertical port grid, so there is no die-scale horizontal data-collection network; and (ii) **consumer-provisioned** — we model that the bandwidth exists at the exit surface, not who consumes it. Under these, data egress collapses to a short local vertical hop (roughly uniform across banking choices), so the packing result is close to physical rather than a loose upper bound. The internal selector cost is retained (t_{switch}, $\mathrm{Vol}_{\mathrm{sel}}$).

Deferred: the thermal boundary condition (a pluggable stub returns $P_{\max}$ from an areal R_{th}; the concurrency cap $N_{\mathrm{SA}}^{\mathrm{active}} \leq \Delta T_{\max}/(R_{\mathrm{stack}}\, E_{\mathrm{access}}/t_{\mathrm{cycle}})$ then caps simultaneously-firing sense amps), the power-delivery network, and the lateral address/command distribution (narrow, assumed provisioned).